DATA ANALYSIS BOOTCAMP

Day 2: Hands-On Project

Toronto Airbnb Market Intelligence

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Quick Recap: The Framework

01 SCOPE → 02 EXPLORE → 03 ANALYZE → 04 COMMUNICATE

Today's Plan

1.	**Load & Profile the Data**	Python / BigQuery	20 min
2.	**Clean & Transform**	Python / BigQuery	20 min
3.	**Analyze: Answer 4 Questions**	Python / BigQuery	25 min
4.	**Build a Tableau Dashboard**	Tableau Public	25 min

THE BRIEF

You're a data analyst at a property management startup expanding into Toronto's short-term rental market.

Leadership needs a market intelligence report before committing resources.

1. Where is the supply concentrated?
2. Where is demand strongest?
3. Professional operators vs casual hosts?
4. How does licensing compliance vary?

Step 1 Load & Profile the Data

Option A: Python (Colab)

```python
import pandas as pd

df = pd.read_csv('listings.csv')

# Shape & types
df.shape
df.dtypes
df.isnull().sum()

# Distributions
df.describe()
df['room_type'].value_counts()
```

Option B: SQL (BigQuery)

```sql
-- Upload CSV to BigQuery
-- Dataset: airbnb_toronto

SELECT
  COUNT(*) AS total_rows,
  COUNT(DISTINCT neighbourhood)
    AS unique_hoods,
  COUNTIF(reviews_per_month
    IS NULL) AS null_reviews
FROM
  airbnb_toronto.listings;
```

What Profiling Reveals

Key findings from the raw data before any transformation

15,776

Total listings

Toronto is a big market

67%

Entire homes

10,575 of 15,776 listings

28

Median min nights

Lots of long-term rentals

3,231

Zero-review listings

20% never booked or reviewed

47%

Multi-listing hosts

7,418 listings from professional operators

57%

Licensed

8,933 of 15,776 have a license

Step 2 Clean & Transform

Fill nulls in reviews_per_month

3,231 listings have NULL. Fill with 0 (no reviews = no observed demand).

Categorize minimum_nights

Create rental_type: Short-term (1-7), Medium (8-27), Long-term (28+). This is a key segmentation axis.

Flag professional hosts

calculated_host_listings_count > 1 = professional operator. This splits casual vs commercial.

Flag licensed listings

license IS NOT NULL = licensed. Regulatory compliance varies by neighbourhood.

Filter to active listings

number_of_reviews > 0 OR availability_365 > 0. Remove ghost listings that skew the data.

STEP 3

ANALYZE

Answer the 4 questions with code and data

Q1 Where is the supply concentrated?

Python

```python
# Top 10 neighbourhoods by listing count
top_hoods = df.groupby(['neighbourhood', 'room_type']).size()
    .reset_index(name='count')
    .sort_values('count', ascending=False)

# Room type distribution
df['room_type'].value_counts(normalize=True).round(3)
```

What you'll find

- Waterfront Communities alone has 2,659 listings (17% of all supply).
- Top 10 neighbourhoods account for ~40% of total supply.
- 67% entire homes, 33% private rooms, <1% shared/hotel.

Q2 Where is demand strongest?

Python

```python
# Demand proxy: avg reviews_per_month by neighbourhood
demand = df[df['number_of_reviews'] > 0].groupby('neighbourhood').agg(
    avg_rpm=('reviews_per_month', 'mean'),
    median_rpm=('reviews_per_month', 'median'),
    total_reviews=('number_of_reviews', 'sum'),
    listing_count=('id', 'count')
).sort_values('avg_rpm', ascending=False).head(15)
```

What you'll find

- High supply ≠ high demand. Some dense areas have low review velocity.
- reviews_per_month is the best demand proxy (normalizes for listing age).
- Compare total volume vs per-listing intensity to separate saturated from underserved.

Q3 Professional operators vs casual hosts?

Python

```
# Professional = host with more than 1 listing
df['host_type'] = df['calculated_host_listings_count'].apply(
    lambda x: 'Professional' if x > 1 else 'Casual')

# Compare performance
df.groupby('host_type').agg(
    listings=('id','count'), avg_rpm=('reviews_per_month','mean'),
    avg_avail=('availability_365','mean')).round(2)
```

What you'll find

- 47% of listings come from multi-listing hosts (7,418 listings).
- Professional operators dominate entire-home supply. Casual hosts are mostly private rooms.
- Top host operates 115 listings. Market is more commercial than it appears.

Q4 How does licensing compliance vary?

Python

```
df['is_licensed'] = df['license'].notna() & (df['license'] != '')

compliance = df.groupby('neighbourhood').agg(
    total=('id','count'),
    licensed=('is_licensed','sum')
)
compliance['pct'] = (compliance['licensed']/compliance['total']*100).round(1)
```

What you'll find

- Overall licensing rate: ~57% (8,933 of 15,776).
- Compliance varies hugely by neighbourhood. Some areas are 80%+, others under 30%.
- Professional hosts tend to be more licensed than casual hosts (they have more to lose).
- Low-compliance + high-supply = regulatory risk zone for new entrants.

Step 4 Build the Dashboard

Export your cleaned data to CSV, connect Tableau Public

Sheet 1: Supply Map

Horizontal stacked bar

Bar chart: Top 15 neighbourhoods by listing count, colored by room type. Shows where supply is concentrated and what type dominates each area.

Sheet 2: Demand vs Supply

Scatter with size encoding

Scatter plot: X = listing count per neighbourhood, Y = avg reviews_per_month. Size = total reviews. Identifies underserved high-demand areas.

Sheet 3: Host & Compliance

Treemap + highlight table

Two side-by-side views: (1) Professional vs Casual host split by room type, (2) Compliance rate by top neighbourhoods as a highlight table.

Your Deliverable

Structure your findings like a real analyst would

1. **Executive Summary (2 sentences)**
   Toronto's Airbnb market is dominated by entire-home listings concentrated in Waterfront + downtown, with nearly half the supply controlled by professional operators. Licensing compliance is 57% citywide but drops below 30% in key growth neighbourhoods.

2. **3 Key Findings (with evidence)**
   Each finding = one sentence + one chart from your dashboard. Don't explain methodology.

3. **1 Recommendation**
   Based on the data, what should the company do? Which neighbourhoods, which property types, which host strategy? Be specific.

4. **Gaps & Next Steps**
   What this data can't tell you: pricing, booking revenue, guest demographics. This builds credibility.

You just did real data analysis.

Not a tutorial. Not a walkthrough.

A question, a dataset, and a deliverable.

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Put this project on GitHub. Put the dashboard on Tableau Public.

That's a portfolio piece.